

Comparing Groups with Two-Way Tables

Reading cells and totals, turning counts into row fractions, and comparing two groups

Directions

- Each cell of a two-way table counts the people in *both* categories at once. The last row and last column hold the totals.
- To compare two groups fairly, turn each count into a fraction or a percentage *of its own row total*. Row counts only compare fairly when the row totals are the same.
- Give fractions in simplest form. Where a question asks for a percent, write the number without the percent sign.

1. Sixty students were asked which sport they play and whether they walk to school.

	Walks	Does not walk	Total
Football	18	12	30
Basketball	10	20	30
Total	28	32	60

How many students play basketball and do not walk to school?

Answer: _____

2. The same survey of sixty students.

	Walks	Does not walk	Total
Football	18	12	30
Basketball	10	20	30
Total	28	32	60

How many students walk to school altogether?

Answer: _____

3. One hundred and forty people were asked whether they prefer cats or dogs.

	Prefers cats	Prefers dogs	Total
Under 30	35	45	80
30 and over	25	35	60
Total	60	80	140

What fraction of the under-30 group prefers cats? Give your answer in simplest form.

Answer: _____

4. The same cats-and-dogs survey.

	Prefers cats	Prefers dogs	Total
Under 30	35	45	80
30 and over	25	35	60
Total	60	80	140

(a) Fraction of the under-30 group preferring dogs = _____

(b) Fraction of the 30-and-over group preferring dogs = _____

(c) Write $<$, $>$, or $=$ between them: (a) _____ (b) _____

5. One hundred students were tracked: whether they attend homework club and whether they passed the end-of-term test.

	Passed	Did not pass	Total
Attends club	42	8	50
Does not attend	30	20	50
Total	72	28	100

What percent of the club attenders passed?

Answer: _____

6. The same homework-club data.

	Passed	Did not pass	Total
Attends club	42	8	50
Does not attend	30	20	50
Total	72	28	100

- (a) What percent of the students who do not attend the club passed? _____

- (b) Describe whether attending the club and passing appear to be associated, using both percentages.

7. One hundred and twenty students recorded how they travel to school.

	Bus	Walk	Cycle	Total
Year 7	24	30	6	60
Year 8	18	24	18	60
Total	42	54	24	120

Which way of travelling is used by the most students overall?

Answer: _____

8. The same travel survey.

	Bus	Walk	Cycle	Total
Year 7	24	30	6	60
Year 8	18	24	18	60
Total	42	54	24	120

(a) Fraction of Year 8 who cycle = _____

(b) Fraction of Year 7 who cycle = _____

(c) Write $<$, $>$, or $=$ between them: (a) _____ (b)

9. One hundred and twenty students tasted a new school lunch and said whether they liked it.

	Liked it	Disliked it	Total
Year 7	45	15	60
Year 9	24	36	60
Total	69	51	120

What percent of all the students surveyed liked the new lunch?

Answer: _____

10. The same school-lunch survey.

	Liked it	Disliked it	Total
Year 7	45	15	60
Year 9	24	36	60
Total	69	51	120

(a) Percent of Year 7 who liked it _____ (b) Percent of Year 9 who liked it _____

(c) Explain whether year group and opinion appear to be associated, and say what the two percentages show that the raw counts on their own would not.
